HDOC Day-47

Question: Write a C program using a variable-length sorted integer array and pointers. Create a function **ceil_index()** to find the index of the first (smallest) element greater than or equal to the target. Print the element and its index, or -1 if no such element exists.

C Program:

```
#include <stdio.h>

int ceil_index(int *arr, int n, int target) {
    int i;
    for (i = 0; i < n; i++) {
        if (*(arr + i) >= target) {
            printf("Ceil element: %d\n", *(arr + i));
            printf("Ceil index: %d\n", i);
            return i;
        }
    }
    printf("-1\n");
    return -1;
}

int main() {
    int n, i, target;
    printf("Enter size of array: ");
    scanf("%d", &n);
    if (n <= 0) {
        printf("-1\n");
        return 0;
    }
    int arr[n];
    printf("Enter sorted array elements:\n");
    for (i = 0; i < n; i++) {
        scanf("%d", &arr[i]);
    }
    printf("Enter target: ");
    scanf("%d", &target);
    ceil_index(arr, n, target);
    return 0;
}
```

Sample Output:

```
Enter size of array: 7
Enter sorted array elements:
5 7 7 8 8 10 12
Enter target: 7
Ceil element: 7
Ceil index: 1
```

Explanation: The array is passed to ceil_index() using its address. The function checks elements from left to right using pointer arithmetic. The first element that is greater than or equal to the target is the ceil element. Checking from the beginning also gives the first index when values repeat. If no element qualifies, the program prints -1.